

# Progression notes

Working log — last updated 2026-09-22. For the project overview, setup, and run commands, see the [README](#). I add to this page as the project grows; the final report will be a trimmed subset of it.

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## Current progress

### Implemented

- **Bilinear reference reader:** A full-resolution sampler that every later representation is read through, so quality differences come from the representation and not the sampling code.
- **S3TC/DXT1 baseline:** Classic 4×4 block compression from scratch. It gives a fixed 6× reduction and lands around 41–43 dB, which is the bar the learned version has to beat.
- **Seam-aware S3TC:** A cross-boundary seam metric plus three endpoint-selection variants (error diffusion, boundary-weighted refit, seam-energy ICM). ICM also raises PSNR. All keep the fixed 6×.
- **Learned representation:** A multi-resolution feature grid decoded by a small MLP, fit per texture, with a training loop (fixed texel set, random minibatches, Adam). Wrapped behind the same interface as the block methods.
- **Full comparison:** all three sizes on all three textures (five seeds each), with PSNR-over-training and size-vs-quality plots and S3TC overlaid.
- **Quantization:** post-training 8-bit quantization of the grid (and optionally the MLP), roughly halving to quartering the stored size for at most a couple of dB.
- **Sourced textures:** eight CC0 textures from ambientCG spanning frequency content, tested with all three sizes and S3TC. S3TC wins on every one, and the gap grows with high-frequency detail; the three most interesting cases are analyzed below.

Every method now exposes the same `compress(texture) / sample(u, v)` interface, so they can be compared on one size-vs-quality axis.

## To implement next

- Trim this progression doc into the final report.

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## The shared read convention

A texel's center sits at  $(i + 0.5) / \text{size}$ , and a coordinate  $u$  maps into continuous texel space as  $x = u * \text{width} - 0.5$ , likewise for  $v / y$ . Lookups outside the texture clamp to the border (replicate padding), matching PyTorch's `F.grid_sample(..., align_corners=False, padding_mode="border")`.

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## Bilinear sampling

Given  $(u, v)$ , find the four surrounding texels, take the fractional position  $(s, t)$  inside the cell, and blend:

$$\text{color}(u, v) = (1-s)(1-t) \cdot c_{00} + s(1-t) \cdot c_{10} + (1-s)t \cdot c_{01} + s \cdot t \cdot c_{11}$$

The four corners are named for their offsets:  $c_{00}$  is the top-left of the cell and  $c_{11}$  the bottom-right. It's vectorized over a batch of coordinates, so the same function handles a few probes or a full megapixel pass.

Correctness validated by:

- Sampling an image at its own texel centers reproduces it exactly (PSNR =  $\infty$ ), which pins down the texel-center convention.
- On random coordinates it matches `F.grid_sample` (`align_corners=False, padding_mode="border"`) to floating-point tolerance, borders included.

## S3TC / DXT1 block compression

S3TC splits the image into 4×4 blocks and spends a fixed 64 bits per block:

- Two endpoint colors, each a 16-bit RGB565 value ( $2 \times 16 = 32$  bits).
- Sixteen 2-bit indices, one per texel ( $16 \times 2 = 32$  bits).

In total, it takes 8 bytes per 16 texels, or 4 bits per texel. This is **6× smaller** than 24-bit RGB. A four-color palette is interpolated from the endpoints, and each texel stores the index of its nearest palette color:

$$c_2 = (2c_0 + c_1) / 3 \quad c_3 = (c_0 + 2c_1) / 3$$

When decoding, the `sample(u, v)` path decodes the image once and then bilinearly samples it with the reader above, so S3TC is read exactly like every other representation.

### Picking the endpoints

The endpoints dominate block quality, so I choose them per block in two steps:

1. **Principal color axis (PCA).** Subtract the block mean, approximate the dominant direction of variation with a few power iterations on the 3×3 covariance, project the 16 texels onto that axis, and take the two extremes. This lines the palette up with the direction the block actually varies in.
2. **Least-squares refit.** Assign each texel to its nearest palette entry, then solve for the endpoints that best reconstruct those colors. Palette entries are fixed blends of the endpoints (`c2` and `c3` above), so each block is a tiny 2-variable least-squares problem with a closed form. Repeat this assign-and-refit twice.

Flat blocks, where the fit is underdetermined, just keep their previous endpoints instead of dividing by a near-zero determinant. Final endpoints are quantized to RGB565, and blocks are always written in the 4-color mode with the larger code first.

## RGB565, and why the divide-by-levels shortcut isn't exact

Each endpoint channel is quantized by multiplying by its level count, rounding, and dividing back:

```
levels = (31, 63, 31)      # 5, 6, 5 bits for R, G, B
q      = round(c * levels)  # integer 5-6-5 code
c_hat  = q / levels         # back to [0, 1]
```

This is close to, but not exactly, how hardware maps a 5- or 6-bit code back to a value. Hardware expands a code by *bit replication* into 8 bits, so the true value would be  $\text{round}(q \cdot 255/31)/255$ . The shortcut implemented instead returns  $q/31$ . Since  $255/31$  isn't an integer, bit replication lands on one of 256 fixed 8-bit levels, while  $q/\text{levels}$  can fall between them. The difference is under a  $1/255$  step and both maps are monotonic with the same half-step rounding error, so the difference is not significant.

**Why does green get the extra bit?** The 5-6-5 split puts six bits on green, one more than red or blue. This is because green carries most of the human eye's perceived luminance (which is probably because sunlight is mostly green which is because green is between blue and red in terms of light wavelength). The Rec. 709 weights are about 0.21 R, 0.72 G, 0.07 B, so a green error is 3x as visible as the same error in red and 10x as visible as one in blue. Spending the extra level on green therefore removes the most visible banding per bit. From what I've seen, every common 16-bit RGB layout favors green because of this.

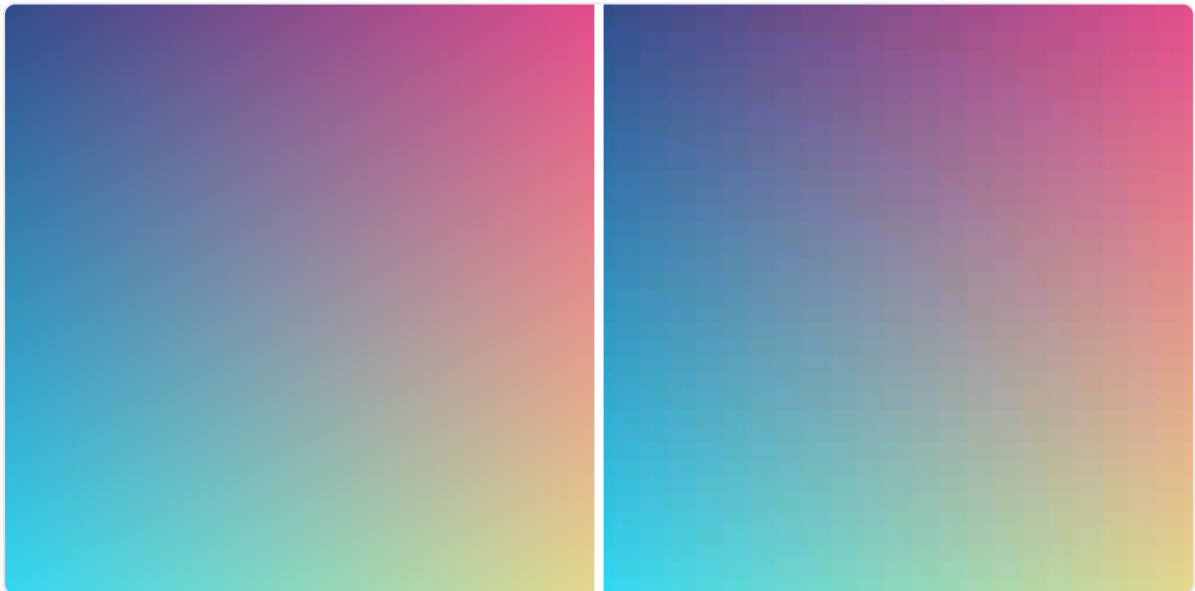
## Results

S3TC is fixed-ratio, so all three textures compress to exactly the same size. PSNR is measured on the decoded image against the original, both normalized to  $[0, 1]$ , with  $\text{PSNR} = -10 \cdot \log_{10}(\text{MSE})$ .

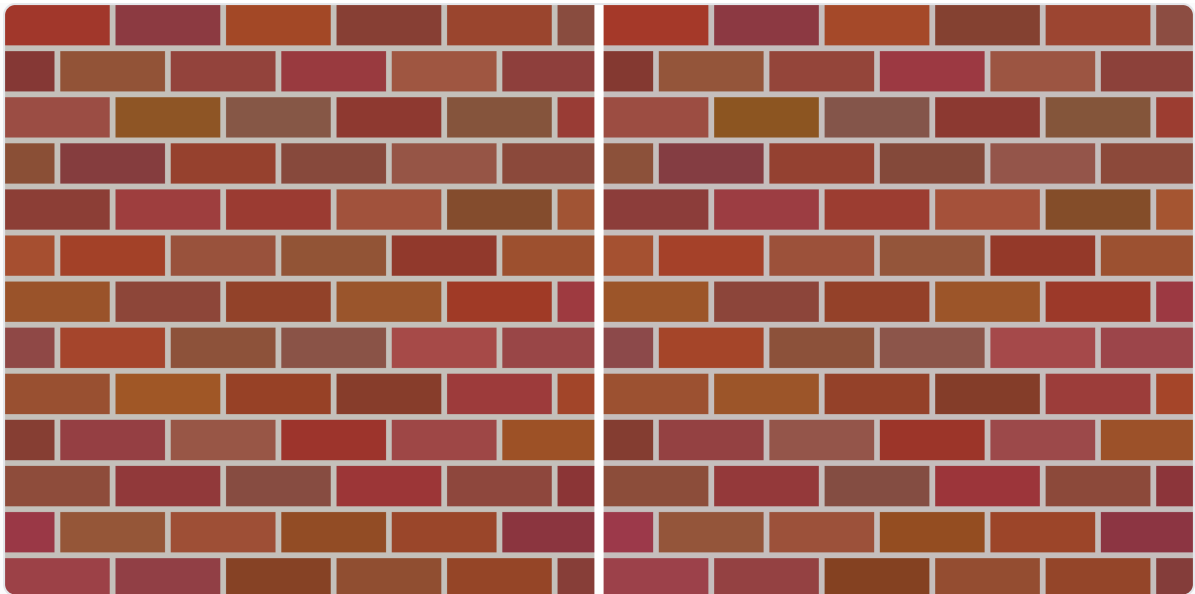
| Texture | PSNR<br>(dB) | S3TC<br>size | Original<br>size | Compression   |
|---------|--------------|--------------|------------------|---------------|
|         | 42.89        | 128 KB       | 768 KB           | 6.00× smaller |

| Texture                          | PSNR<br>(dB) | S3TC<br>size | Original<br>size | Compression   |
|----------------------------------|--------------|--------------|------------------|---------------|
| gradient (smooth, low-frequency) |              |              |                  |               |
| bricks (structured, sharp edges) | 41.39        | 128 KB       | 768 KB           | 6.00× smaller |
| clouds (fractal, multi-scale)    | 42.63        | 128 KB       | 768 KB           | 6.00× smaller |

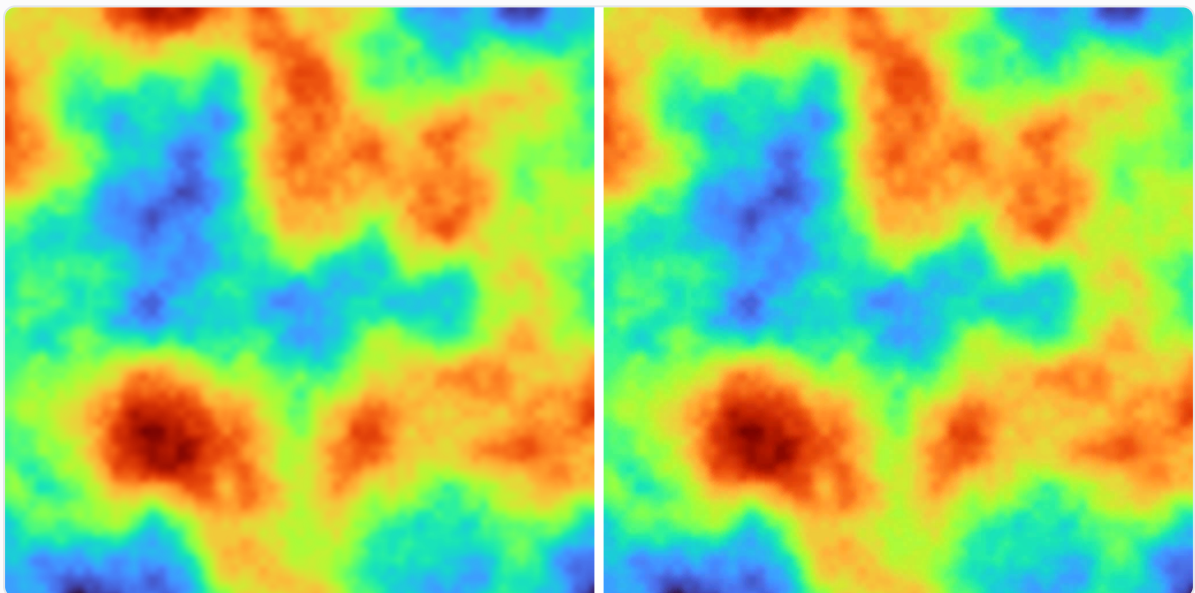
"Compression" is raw size divided by compressed size (a 6× reduction); the compressed file is one sixth of the original.



Gradient: original (left) vs. S3TC (right), 42.89 dB. The smooth ramp is the hardest case for a block format — the reconstruction shows faint block stepping.



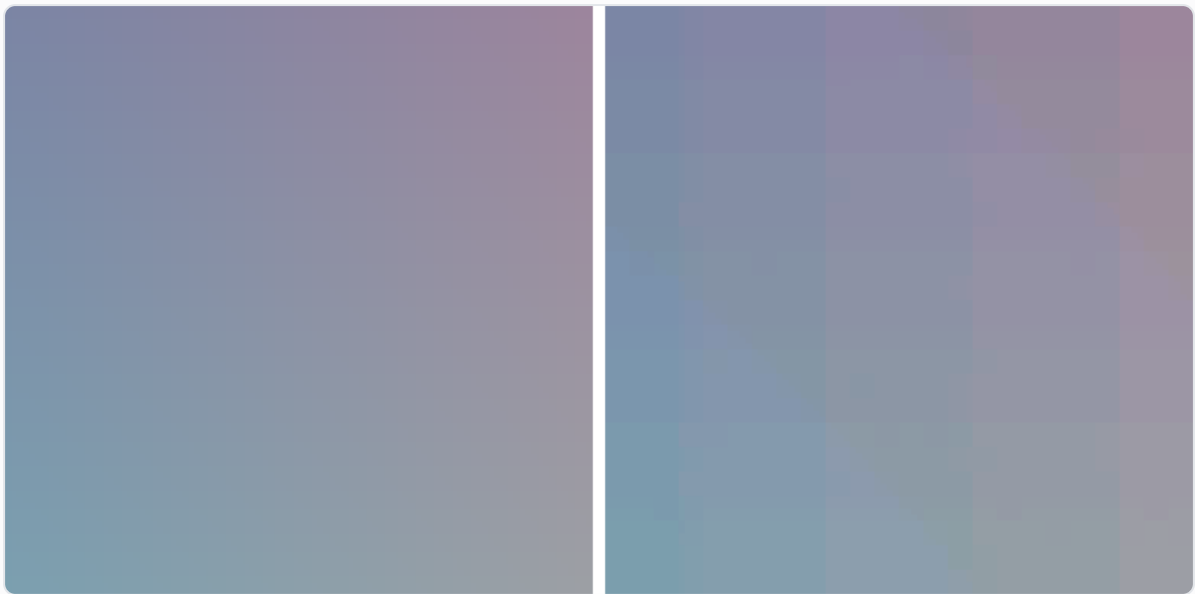
Bricks: original (left) vs. S3TC (right), 41.39 dB. Brick faces and mortar survive, but their colors shift slightly inside each block.



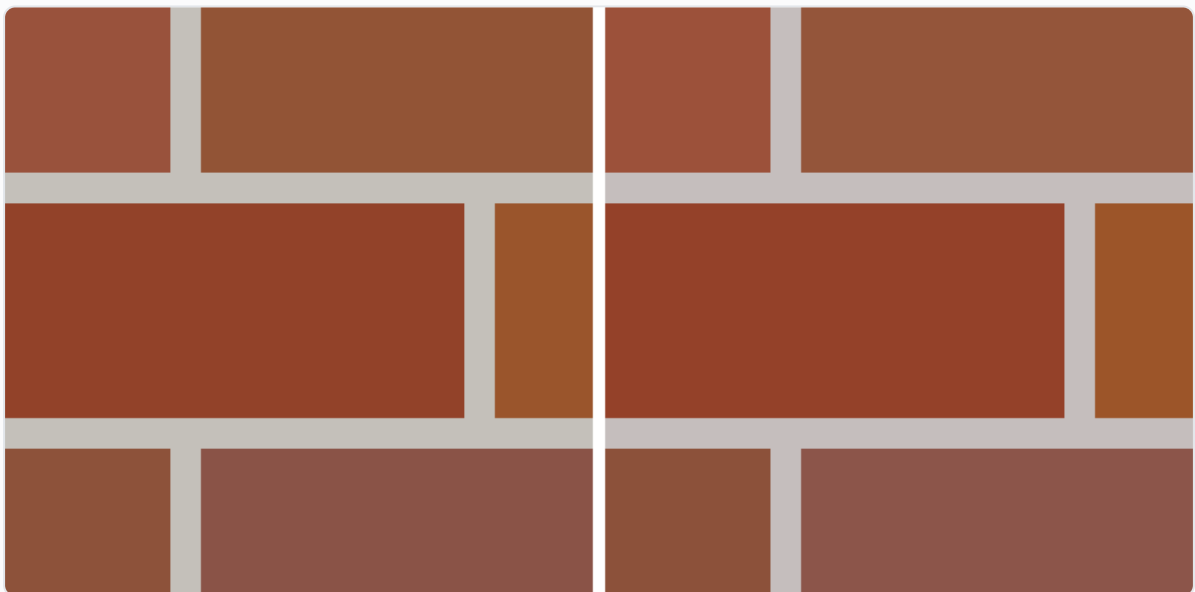
Clouds: original (left) vs. S3TC (right), 42.63 dB. The soft, low-contrast detail mostly survives, with subtle blocking.

### **S3TC is sometimes not good**

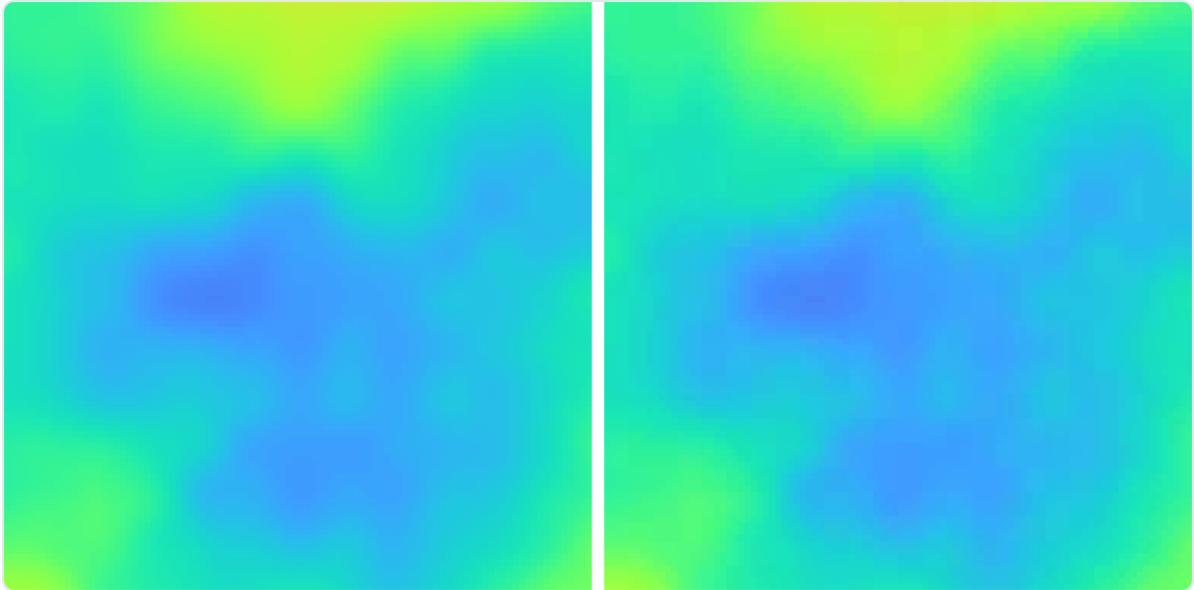
The clearest failure is smooth gradients. The crops below are 96×96 texels, nearest-neighbor upscaled 4×, so each 4×4 block becomes a 16×16 square. Figures are regenerated by `scripts/make_figures.py`.



Gradient: the original ramps smoothly; S3TC shows block-to-block banding.



Bricks: flat brick color picks up blocky shading from the palette fit.



Clouds: soft detail survives; faint blocking in low-contrast areas.

This behavior is because of the nature of S3TC and cannot really be improved by the per-block fit alone.

- The palette is four points on the segment between two endpoints. A smooth image varies in two dimensions inside a block, but the block can only represent colors near that 1-D line, so the leftover error shows up as a color step at every block boundary, causing the banding.
- High-frequency detail that doesn't align with the 4×4 grid can't be represented; edges get smeared or stepped.
- Every block costs the same 64 bits whether it's flat or busy, and nothing is shared between blocks.
- RGB565 rounds each endpoint to 5 or 6 bits before the palette is even built, adding a second error source on top of the palette fit.

At this stage the baseline is 128 KB (6×) at roughly 41–43 dB. A learned representation will replace the fixed palette with a shared feature grid plus an MLP, so its size will be tunable instead of fixed. Because of this, I will compare them by matching size then comparing PSNR and matching PSNR then comparing size.

## Can the S3TC endpoint choice reduce the blocking issues?

The 4-colors-per-block limit is structural, but the baseline picks endpoints block by block with no idea what its neighbors chose, so it leaves recoverable seam on the table. I kept the bit budget and the decoder exactly the same and changed only how the endpoints are chosen, trying three neighbor-aware or global approaches in increasing cost.

### Seam metric

To measure how bad/visible seams were, I made a seam error metric. For every pair of adjacent texels straddling a block boundary, compare the reconstruction's color step to the original's:

$$e = \text{mean } | (\text{rec}[a] - \text{rec}[b]) - (\text{orig}[a] - \text{orig}[b]) |$$

- Subtracting the original's own gradient means a perfect reconstruction scores 0 even on a textured image, and only added discontinuities count.
- Normalized color units, lower is better. It is not identical to perceived blocking, but it should be well-correlated.
- Baseline: gradient 0.00225, clouds 0.00690, bricks 0.00059.
- There is a floor: RGB565 rounding alone contributes about half the seam. Encoding with float endpoints drops the gradient seam from 0.00225 to 0.00119, so endpoint quantization is a real part of the artifact.

### Endpoint tuning

- Instead of selecting endpoints from scratch using a new selection function, I first ran the local PCA-based S3TC endpoint selector (the standard one) and then created loss functions that took into account the blocking between a block's neighbors and ran gradient descent-like algorithms with these loss functions achieve incremental improvement. I experimented with a few different loss functions and ways of iterating through and incrementing endpoints. Some worked better than others.

### Attempt 1: Block error diffusion

- **Idea:** a block's reconstruction error on its right and bottom edges is fed forward into the target colors of the block to its right or below, before that block is fit, so errors cancel across the seam instead of stacking up.
- **Tuning Hyperparameters:** the seam falls as alpha drops and bottoms out around 0.1 (mean seam  $\times 0.94$  across gradient and clouds); by 0.75 it adds seam instead of removing it. PSNR moves the other way, rising with alpha until about 0.5 (+0.28 dB). It converges within two passes, so I set alpha 0.1 and stop early.
- **Results:** gradient seam  $-8\%$  and +0.18 dB, clouds seam  $-4\%$  and +0.14 dB.

### Attempt 2: Boundary-weighted refit

- **Idea:** refit each block's edge texels toward `neighbor_decoded + original_step`, so the block's step across the seam matches the original's, and weight edge texels higher. This targets the seam metric directly.
- **Tuning Hyperparameters:** gamma 0.5 to 0.75 minimizes the seam (mean  $\times 0.91$ ); raising the boundary weight to 2 trims a little more but starts costing PSNR, so I leave it at 1.0. It converges after one or two passes.
- **Results:** best seam on the gradient ( $-13\%$ ) with a small PSNR gain. On clouds it trims seam  $-5\%$  but gives up 0.1 dB.

### Attempt 3: Seam-energy search (ICM)

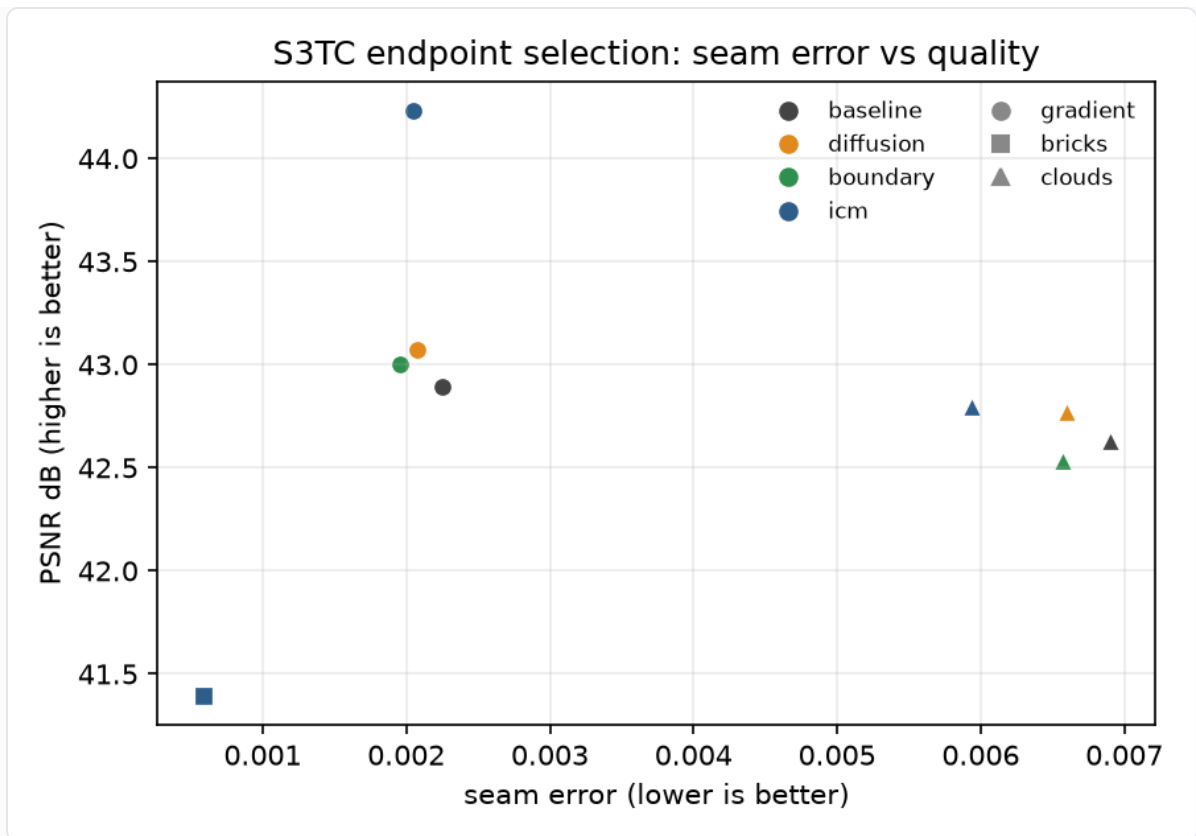
- **Idea:** iterated conditional modes. Each block tries candidate endpoint pairs (its current pair, the PCA extremes, the baseline least-squares fit, and blends toward each of its four neighbors), decodes them, and keeps the one minimizing its own squared error plus `lam  $\times$  added-seam`.
- The candidates are evaluated on the *quantized* decode, so the search also finds endpoints that survive RGB565 rounding, which is why PSNR improves too.
- **Tuning Hyperparameters:** lam sets the trade-off. Low lam (0.1 to 0.5) maximizes PSNR (+1.1 dB) while still cutting seam; lam 1 to 2 is the seam sweet spot (mean  $\times 0.885$ ); lam 10 and above over-constrains the clouds and PSNR collapses. beta barely moves the seam but higher values favor smooth textures (up to +2.3 dB on the gradient at beta 0.8) at a small cost on clouds,

so 0.5 stays balanced. Seam keeps improving out to 8 to 12 passes; 4 captures most of it.

- **Results:** best overall. Gradient +1.34 dB and seam –9%; clouds +0.16 dB and seam –14%.

## Results

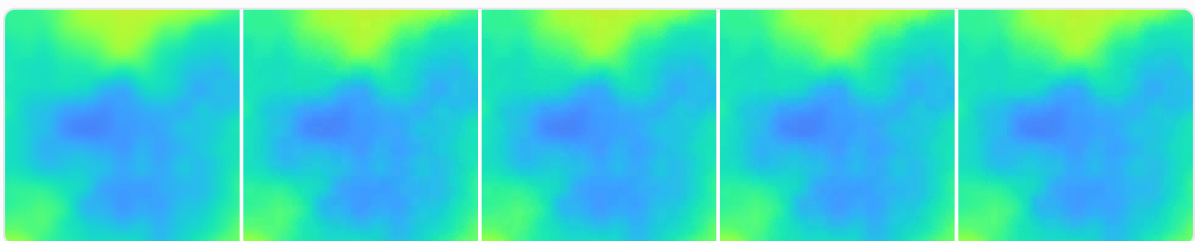
| Texture  | Method      | PSNR (dB) | Seam error |
|----------|-------------|-----------|------------|
| gradient | baseline    | 42.89     | 0.002247   |
| gradient | diffusion   | 43.07     | 0.002072   |
| gradient | boundary    | 43.00     | 0.001959   |
| gradient | ICM         | 44.23     | 0.002046   |
| clouds   | baseline    | 42.63     | 0.006901   |
| clouds   | diffusion   | 42.77     | 0.006595   |
| clouds   | boundary    | 42.53     | 0.006575   |
| clouds   | ICM         | 42.80     | 0.005936   |
| bricks   | all methods | 41.39     | 0.000585   |



PSNR against seam error. Up and to the left is better. ICM (blue) is highest, boundary (green) is leftmost on the gradient, and the clouds points sit furthest right because that texture has the most seam to begin with.



Gradient, 4x zoom. Left to right: original, baseline, diffusion, boundary, ICM. The banding gets progressively less pronounced, most visibly in the ICM panel.



Clouds, 4x zoom, same order. ICM removes most of the visible block structure in the low-contrast areas.

## Take aways

- **ICM (method 3) is the best all-rounder:** the only method that improves PSNR on every non-flat texture and cuts the seam the most on clouds, for a few passes of extra compute and no change in stored size.
- **Boundary refit (method 2) is the cheapest seam win** and wins outright on the smooth gradient, but it can trade a little PSNR for the seam on busier textures.
- **Error diffusion (method 1) is a mild, safe nudge** only at low strength.
- **Bricks is unchanged by all three.** Its blocks are flat, so the baseline already has essentially no added step to fix. A useful sanity check that the methods only touch recoverable seam.
- **Quantization is the remaining floor.** None of these remove the roughly half of the seam that comes from RGB565 rounding. Coordinating the endpoint quantization itself, or searching the RGB565 lattice globally, is probably the best next move to reduce blocking.
- All of this is free in stored size: the representation is still exactly 8 bytes per block (a fixed 6×), only the endpoint choice changed.

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## Neural representation

Instead of using a fixed palette as in S3TC, a multi-resolution feature grid and MLP is used to turn blended features into an RGB color.

### Feature grid

- One learnable `(1, feat_dim, R, R)` tensor per resolution (e.g. 16, 32, 64, 128).
- A lookup bilinearly blends the four surrounding feature vectors and concatenates across levels, so `L` levels of width `feat_dim` give `L × feat_dim` channels. Coarse levels carry broad structure, fine levels carry detail.
- The lookup uses the *same* texel-center convention as the full-resolution sampler (`grid_sample`, `align_corners=False`, border clamping). A

single 3-channel level reproduces the sampler to 1e-7, so quality differences come from the representation, not from how the texture is read.

## Decoder

- Two 64-wide ReLU hidden layers, then a sigmoid head for RGB in [0, 1].
- The decoder is very small (about 4.5k to 5.4k weights) compared to the grid, so almost all of the storage costs for this representation come from the feature grid.

## Training

- The training set is every texel center of the image, and each step draws a random 16,384-texel minibatch.
- Trained via adaptive moment estimation (Adam) at 1e-2, 2000 steps, MSE loss. Minimizing loss also happens to be maximizing PSNR, since they are directly correlated with  $\text{PSNR} = -10 \cdot \log_{10}(\text{loss})$ .

## Sizes

| Model  | Levels            | feat_dim | features | params | KB (f32) | vs raw |
|--------|-------------------|----------|----------|--------|----------|--------|
| small  | (64)              | 2        | 2        | 12,739 | 49.8     | 15.4×  |
| medium | (16, 32, 64)      | 2        | 6        | 15,555 | 60.8     | 12.6×  |
| large  | (16, 32, 64, 128) | 4        | 16       | 92,483 | 361.3    | 2.1×   |

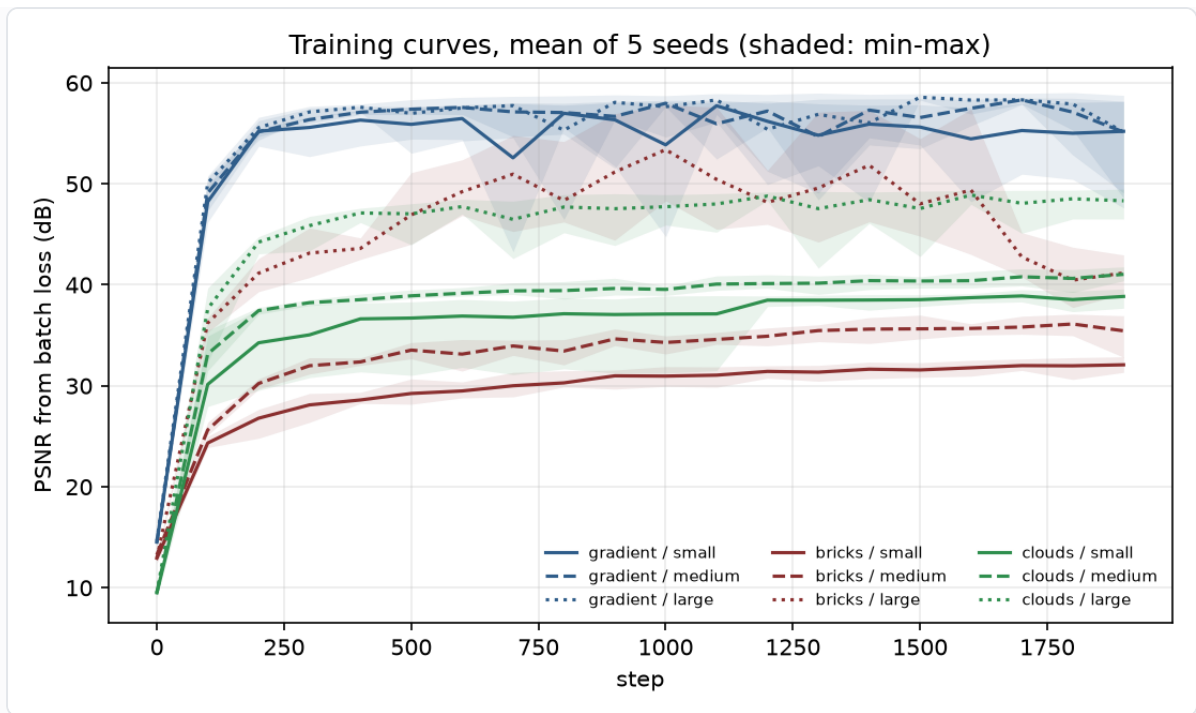
- The three sizes share the same decoder shape but have different feature grids.
- Storage is dominated by the grid, and even the small model is well under S3TC's fixed 6×

## Results

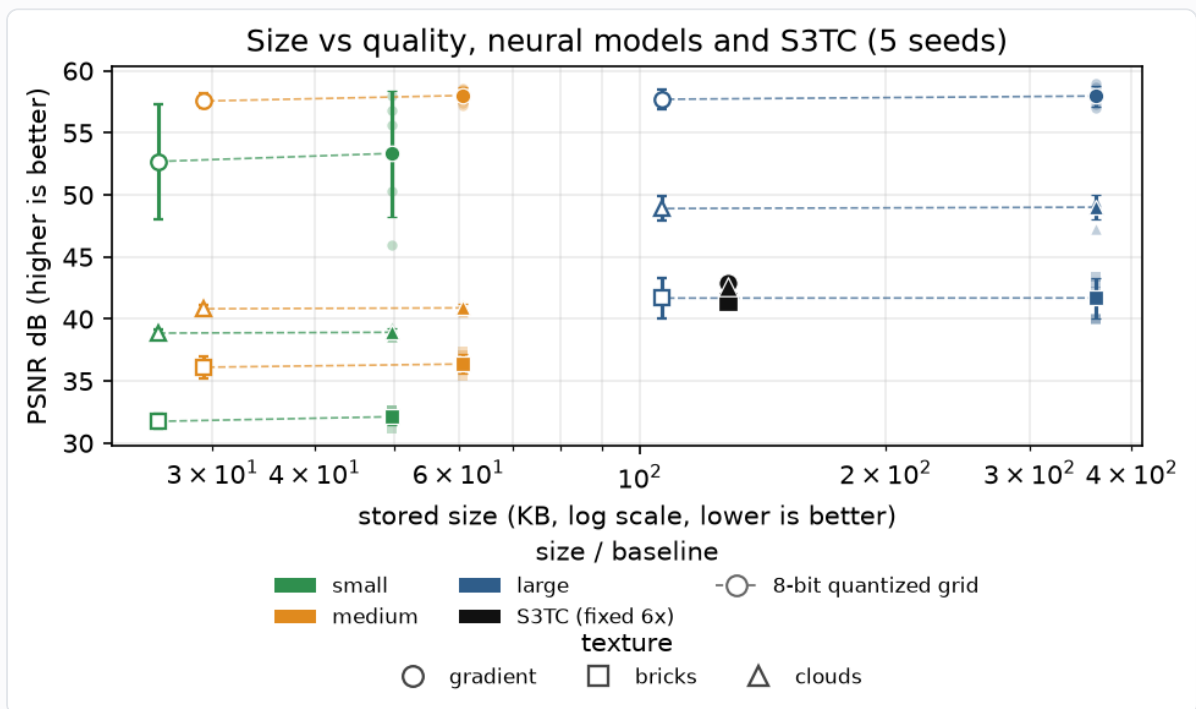
### Size vs quality

All three architectures on each of the three textures, averaged over five seeds (GPU training has a few dB of run-to-run spread). S3TC is the fixed 6× point at 128 KB. The 8-bit column is the grid quantized with the decoder still in float32; see Quantization below for the rest.

| Texture  | Model  | float PSNR | float size (ratio) | 8-bit PSNR | 8-bit size (ratio) |
|----------|--------|------------|--------------------|------------|--------------------|
| gradient | small  | 53.3 ± 5.1 | 49.8 KB (15.4×)    | 52.7       | 25.8 KB (29.8×)    |
| gradient | medium | 58.0 ± 0.6 | 60.8 KB (12.6×)    | 57.6       | 29.3 KB (26.2×)    |
| gradient | large  | 58.0 ± 0.8 | 361.3 KB (2.1×)    | 57.7       | 106.3 KB (7.2×)    |
| gradient | S3TC   | 42.9       | 128.0 KB (6.0×)    |            |                    |
| bricks   | small  | 32.1 ± 0.6 | 49.8 KB (15.4×)    | 31.8       | 25.8 KB (29.8×)    |
| bricks   | medium | 36.4 ± 0.8 | 60.8 KB (12.6×)    | 36.1       | 29.3 KB (26.2×)    |
| bricks   | large  | 41.7 ± 1.6 | 361.3 KB (2.1×)    | 41.7       | 106.3 KB (7.2×)    |
| bricks   | S3TC   | 41.4       | 128.0 KB (6.0×)    |            |                    |
| clouds   | small  | 38.9 ± 0.3 | 49.8 KB (15.4×)    | 38.9       | 25.8 KB (29.8×)    |
| clouds   | medium | 40.9 ± 0.3 | 60.8 KB (12.6×)    | 40.8       | 29.3 KB (26.2×)    |
| clouds   | large  | 49.0 ± 1.0 | 361.3 KB (2.1×)    | 48.9       | 106.3 KB (7.2×)    |
| clouds   | S3TC   | 42.6       | 128.0 KB (6.0×)    |            |                    |



PSNR over training for all nine combinations, mean of five seeds (shaded band is min-max). The large model is both the best and the most stable; the small model saturates early.



Size vs quality. Color is the size or baseline (S3TC is black), marker shape is the texture, and faint dots are the individual seeds. Filled points are float32; open points are the 8-bit quantized grid, joined by a dashed line to show the size saved. S3TC sits at a fixed 6x, 128 KB. Up and to the left is better.

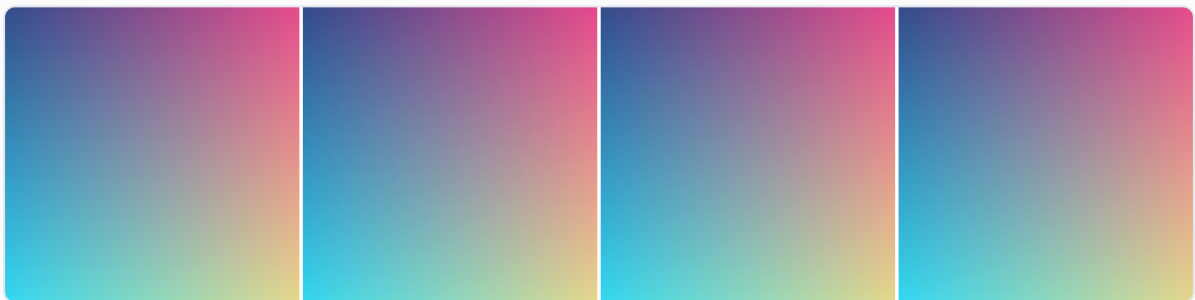
## Quantization

Fitted models are quantized to 8-bit post-training. Each stored array is quantized separately over its own range: round to the nearest of 256 levels, store the index  $q$ , and decode with  $\hat{x} = l_0 + q \cdot \text{scale}$ . That is one byte per value plus the two float32 range endpoints per array. The feature grid is always quantized because it takes up the vast majority of the space, and I tested with and without quantizing the decoder MLP.

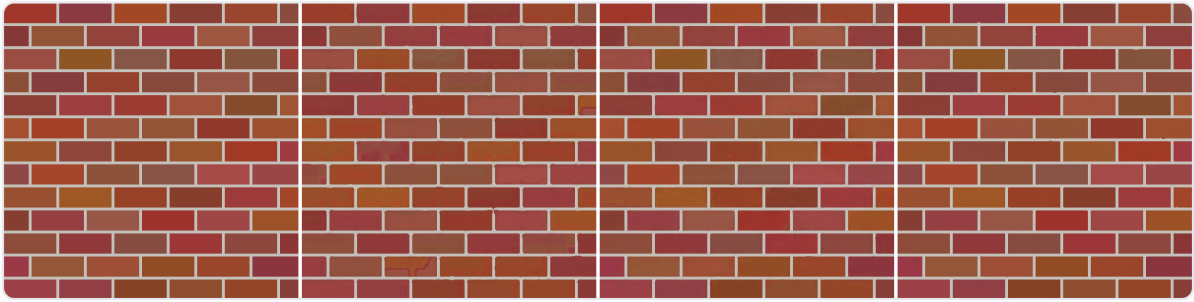
| Model  | float32         | 8-bit grid      | 8-bit grid + MLP |
|--------|-----------------|-----------------|------------------|
| small  | 49.8 KB (15.4×) | 25.8 KB (29.8×) | 12.5 KB (61.5×)  |
| medium | 60.8 KB (12.6×) | 29.3 KB (26.2×) | 15.3 KB (50.3×)  |
| large  | 361.3 KB (2.1×) | 106.3 KB (7.2×) | 90.4 KB (8.5×)   |

- **Quality losses:** quantizing only the grid costs 0.01 to 0.65 dB (the large model loses under 0.3 dB). Also quantizing the decoder costs up to 1.9 dB on the small model, about 0.9 dB on medium and large. The grid features span a narrow range, so a bucket is a tiny fraction of the value scale and the rounding shows up as a small uniform error, not structure. The small model feels the decoder most because its MLP is a larger share of its total size and its weights have a wider dynamic range.
- Quantization makes the neural representation consistently more efficient than the baseline. At full 8-bit quantization, the large model now beats S3TC on PSNR for all three textures while also being smaller, and the small model lands at 12.5 KB (61.5×) with 52.4 / 31.3 / 38.4 dB.

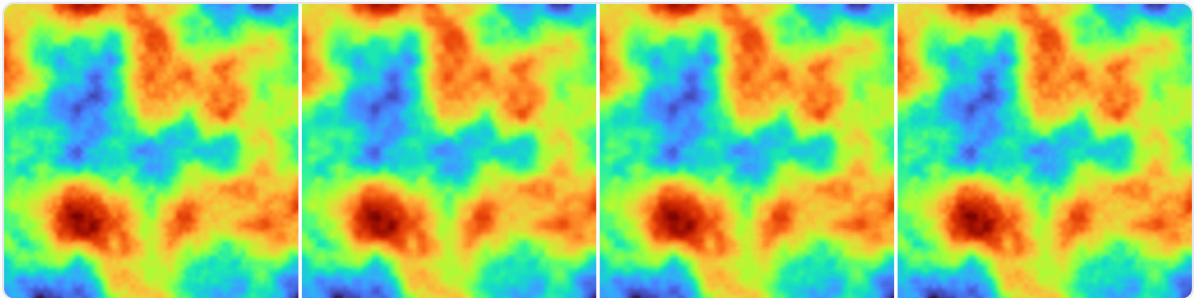
## Reconstructions



Gradient: original, then small, medium, large.



Bricks: original, then small, medium, large. The small model smears the mortar lines; large resolves them.



Clouds: original, then small, medium, large. Fine multi-scale detail only appears at the large size.

### Which architecture wins, and does it beat S3TC?

- **Gradient:** medium quantized seems to be the best PSNR for its size, but all three beat S3TC by about 10 dB or more. With the small model, one unlucky seed stalled near 46 dB, which increased standard deviation by a lot.
- **Bricks:** large quantized performs the best; small collapses to 32 dB because a single 64-level grid cannot hold sharp mortar edges. In float32 the large model only ties S3TC (41.4 dB) while being larger, and for some reason after 8-bit quantization the PSNR improves.
- **Clouds:** large quantized wins by a wide margin, since only it has both coarse and fine levels plus enough features for the fractal detail.
- Transitioning from small to medium (same decoder, extra coarse levels) gains +4.7 dB on gradient, +4.3 on bricks, +2.0 on clouds. Going from medium to large (feat\_dim 2 to 4 plus the 128 level) gains 0 dB on gradient but +5.3 on bricks and +8.1 on clouds. Overall, the busier the texture's frequency content, the more the extra capacity is worth.

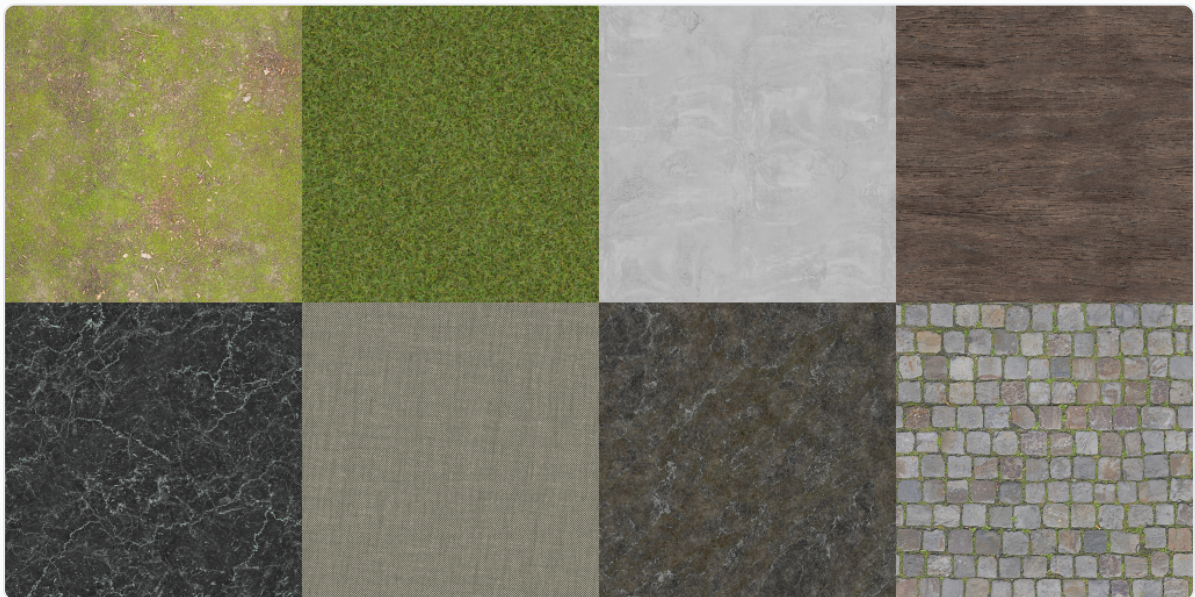
- **Beat S3TC?** The large quantized models are the only ones that are all-around better than S3TC. Without quantization, the large model is bigger than S3TC and the small and medium models often perform equally or worse than S3TC.
- It might be important to note that I just used a fixed 2000 steps and  $lr\ 1e-2$  with no tuning, and GPU training is nondeterministic, so a heavy-tailed config (gradient small/medium) needs several seeds for a stable mean.

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## Sourced textures

The three starter textures turned out to be unusually smooth, so I sourced eight more from ambientCG (all CC0, credited in `assets/own/CREDITS.md`) with deliberately different frequency content and structure: fine stochastic detail (gravel, grass), a fine repetitive weave (fabric), smooth low-contrast (concrete), directional grain (wood), sparse strong veins (marble), multi-scale natural (rock), and structured sharp edges (paving).

To measure frequency content I used the mean absolute Laplacian of the grayscale image, where a higher value means more high-frequency energy. The starter textures scores are 0.006 for the gradient, 0.008 for the clouds, and 0.051 for the bricks, while the sourced set spans from 0.036 to 0.477, meaning the new textures have a lot more high-frequency detail.

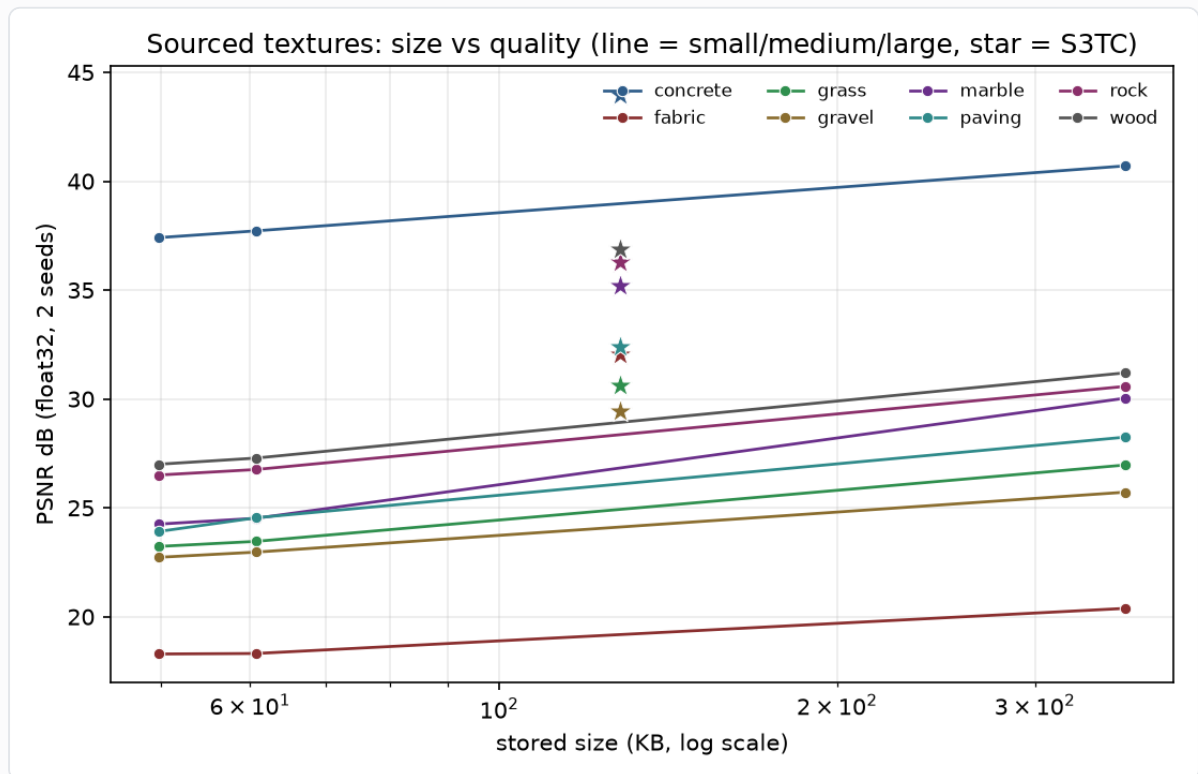


Top row: gravel, grass, concrete, wood. Bottom row: marble, fabric, rock, paving.

## Results

All of the representation sizes are the same as previous tests. I didn't bother quantizing the learned models since their accuracy was already not good enough to beat PSNR.

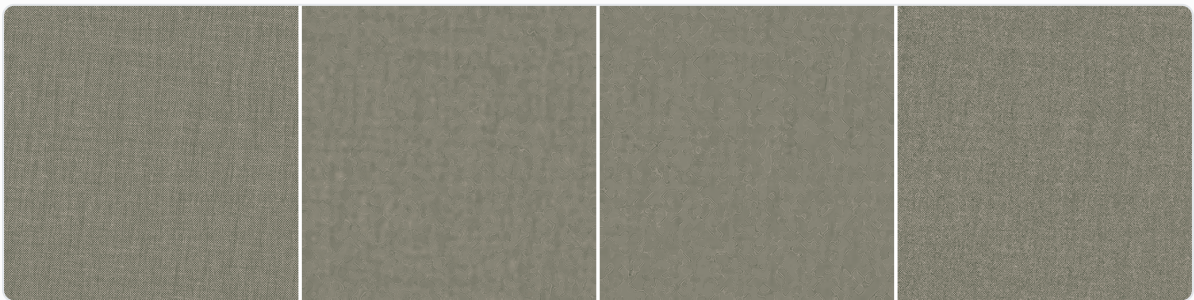
| Texture  | detail | S3TC | small | medium | large |
|----------|--------|------|-------|--------|-------|
| concrete | 0.036  | 44.0 | 37.4  | 37.7   | 40.7  |
| wood     | 0.112  | 36.9 | 27.0  | 27.3   | 31.2  |
| rock     | 0.113  | 36.3 | 26.5  | 26.8   | 30.6  |
| marble   | 0.124  | 35.2 | 24.3  | 24.5   | 30.0  |
| paving   | 0.139  | 32.4 | 23.9  | 24.5   | 28.2  |
| grass    | 0.138  | 30.6 | 23.2  | 23.5   | 27.0  |
| gravel   | 0.178  | 29.4 | 22.7  | 23.0   | 25.7  |
| fabric   | 0.477  | 32.1 | 18.3  | 18.3   | 20.4  |



Each line is small to medium to large for one texture; the star is S3TC. Every S3TC star sits above its texture's whole neural curve, and the gap widens with the detail score.

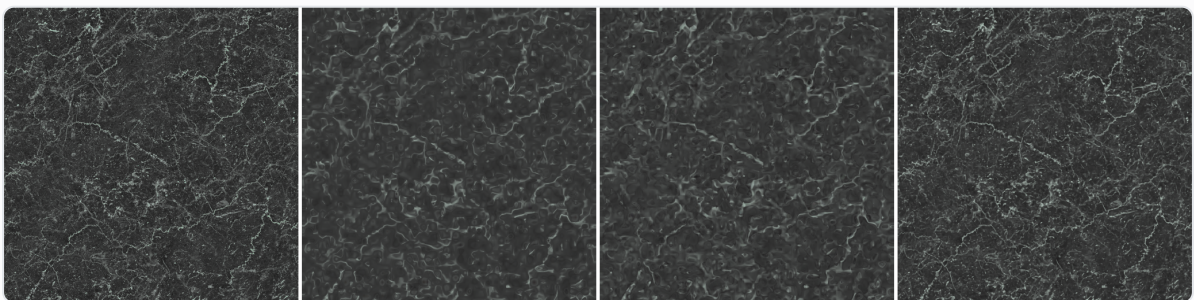
On every sourced texture S3TC beats the neural model, and it beats even the large model while using roughly a third of its storage. Quantization obviously does not improve the performance, but it also surprisingly doesn't make it that much worse either.

### Three interesting results



Fabric: original, then small, medium, large.

- **High frequency textures (Fabric, detail 0.477):** This is a regular woven pattern, which you might expect a small grid to capture, but the weave is near the grid's finest resolution. Instead of dropping the weave, the model invents smooth large-scale blobs that are not in the texture. A bilinear grid is a low-pass filter, and content above its cutoff does not simply disappear, but instead turns into spurious low-frequency structure.



Marble: original, then small, medium, large.

- **Resolution findings (Marble, detail 0.124):** Marble is sparse thin bright veins on a dark, fairly flat base, so almost all the error is on a small fraction of texels. Small and medium cannot resolve the veins and instead blur them into wavy bands. The large is a lot better but it is still not as good as S3TC.



Concrete: original, then small, medium, large.

- **Low-frequency textures still don't do great (Concrete, detail 0.036):** This is the smoothest sourced texture and the closest to the starter gradient texture that the neural model did best on (which had detail 0.006). The neural representation still loses to S3TC by 3 dB. The reconstruction loses the fine aggregate speckle and shallow surface variation, leaving a clean but blurry surface.

## Conclusions

- **Feature grid + MLP is a low-pass representation.** The effective bandwidth is set by the finest grid level and by how much the small MLP can sharpen, making it inferior to S3TC for capturing small details. Detail it cannot carry becomes blur, and just above the cutoff it can become spurious low-frequency structure.
- Feature grid + MLP is preferable when dealing with a wide range of colors and low-frequency details, while S3TC captures full detail but giving up on color range.
- S3TC adapts per block, so it holds edges and fine detail that a global coordinate network smears. Beating it needs the learned model to either match that adaptivity or buy enough resolution and capacity to not need it.
- With the current configuration (small grids, a  $2 \times 64$  MLP, 2000 steps), the feature grid + MLP does not beat S3TC on natural textures with high-frequency details and low color ranges. In the future, this feature grid + MLP implementation might benefit from higher grid resolutions and more levels, more feature channels, positional/frequency features, or a hash encoding so that fine detail is learnable. More aggressive quantization could also be experimented with as the current level of quantization did not have much of an effect on the final representation fidelity.

## Log

2026-09-20 Project scaffold, environment, and CUDA setup. Bilinear sampler with the shared read convention, verified against `grid_sample`. S3TC/DXT1 baseline with PCA endpoints and least-squares refits; 6× fixed at ~41–43 dB. Figures and this page added.

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2026-09-20 Added a cross-boundary seam metric and three seam-aware endpoint-selection variants (block error diffusion, boundary-weighted refit, seam-energy ICM). ICM improves PSNR and cuts the seam the most; boundary refit gives the best seam on smooth gradients; error diffusion is a mild nudge. All keep the fixed 6× size.

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2026-09-21 Added the learned representation: multi-resolution feature grid, decoder MLP, and the training loop. First fit (small, gradient) reaches ~54–58 dB in ~9s at 15.4×, versus S3TC's 42.9 dB at 6×.

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2026-09-21 Full comparison done: 3 sizes × 3 textures × 5 seeds. The neural model wins outright on the smooth gradient (small beats S3TC at a quarter of the size); on bricks and clouds only the large model beats S3TC on PSNR, at 2.1× versus 6×. Added the PSNR-vs-size and training-curve plots and the per-texture reconstructions.

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2026-09-21 Added post-training 8-bit quantization of the grid (and optionally the decoder) and re-ran the comparison. Size drops 2-4× for at most a couple of dB. After quantization the large model beats S3TC on all three textures at a smaller stored size.

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2026-09-21 Sourced eight CC0 textures (ambientCG) spanning frequency content and added them to the provided set. S3TC beats the neural model on all eight, and the gap grows with detail; the starter textures were unusually smooth (detail 0.006–0.051 vs 0.036–0.477). Wrote up fabric (high-frequency failure), marble (resolution-limited), and concrete (even the smoothest sourced texture loses).

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